

Template for Assessment 1

Questions 1

Plot the value of GDP_t for 1960:Q1 to 2002:Q4. Does the inf lt series look stationary (substantiate your answer)?

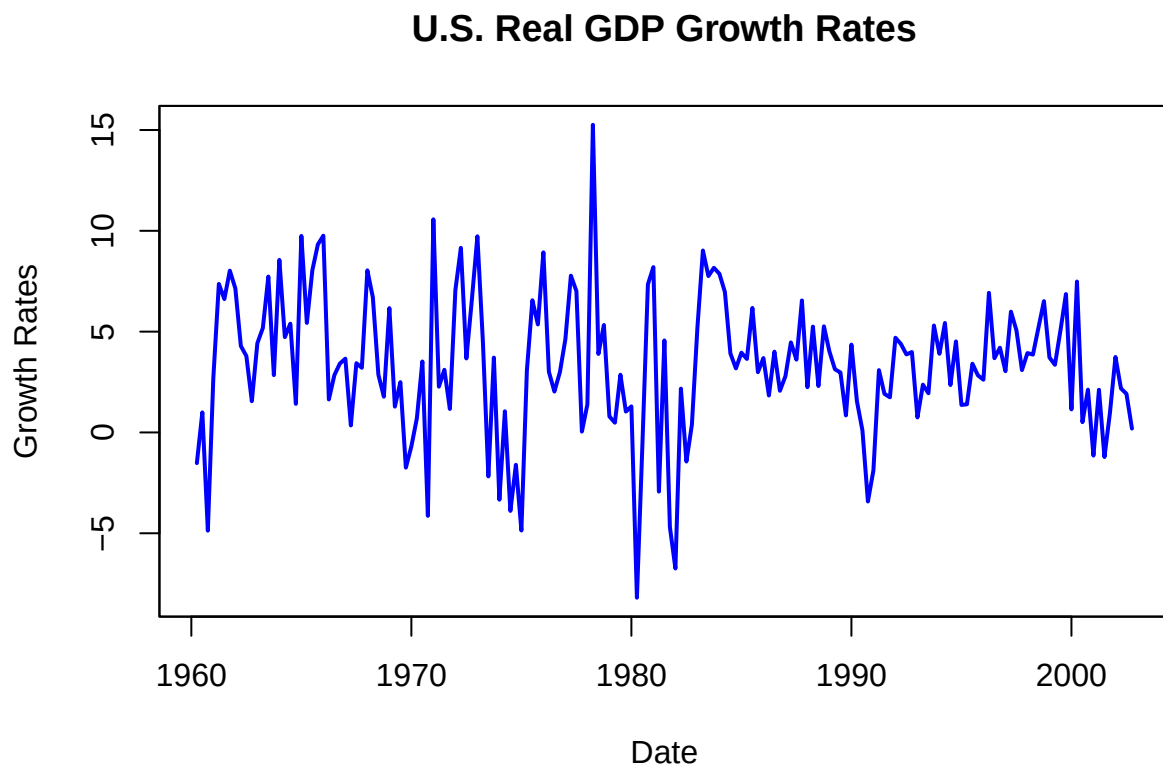


Figure 1: The graph shows U.S. Annual Real GDP Growth Rates for each quarter between 1960 and 2002.

A time series is weakly stationary if its mean and variance remain constant over time and its autocovariance depends only on the lag between observations, not on the specific time at which it is measured. The GDP series does not appear to be stationary. It shows a steady increasing trend from the 1960s to 2002. Additionally, the distribution of GDP between 1960 and 1970 differs from the distribution between 1990 and 2000. The mean of the GDP growth rate distribution also seems to change over time, indicating that the GDP rate is non-stationary.

Warning: This answer is just a template and is based solely on the GDP plot in your lecture. A proper discussion should include three levels of analysis(which I asked in the assignment): the plot level, the autocorrelation function (ACF) level, and a unit root test. Therefore, this type of answer is not sufficient for your assignment.

Questions 2

Estimate an AR(1) model for $GDPGrowth_t$ using the `dynlm()` command.; that is, run an OLS regression of $GDPGrowth_{t-1}$ on $GDPGrowth_t$ using the `dynlm()` command. Report the estimated regression. Does knowing the change in inflation over the current quarter help predict the change in inflation over the next quarter?

Dependent variable: $GDPGrowth_t$		
	Model 1	Model 2
$GDPGrowth_{t-1}$	0.302*** (0.073)	0.302*** (0.078)
Constant	2.349*** (0.353)	2.349*** (0.394)
Observations	170	
R ²	0.092	
Adjusted R ²	0.086	
Residual Std. Error	3.303	
F Statistic	16.955***	

Note: Model 1 uses non-robust standard errors; Model 2 uses HAC standard errors.
*p<0.1; **p<0.05; ***p<0.01

Table 1: AR(1) Model Regression Results for GDP Growth

This table presents the results from an autoregressive AR(1) model estimated using ordinary least squares (OLS), where GDP growth in period t ($GDPGrowth_t$) is regressed on its lagged value ($GDPGrowth_{t-1}$).

The robust standards are reported in the second columns. The coefficient on $GDPGrowth_{t-1}$ is 0.302, statistically significant at the 1% level ($p < 0.01$), indicating a moderate degree of persistence in GDP growth. The constant term is also positive and significant. The model explains approximately 9.2% of the variation in GDP growth, as indicated by $R^2 = 0.092$, with an adjusted R^2 of 0.086. The F-statistic (16.955) suggests that the overall model is statistically significant. Standard errors (reported in parentheses) are non-robust and may not account for heteroskedasticity or autocorrelation.

Since the coefficient on $GDPGrowth_{t-1}$ is statistically significant at all conventional significance levels, we conclude that the knowing the change in GDP growth over the current quarter does help predict the change in GDP growth over the next quarter. This is consistent with the significantly positive autocorrelation of GDP growth.

Warning: *This answer is just a template and is based solely on the GDP Growth data in your lecture. Therefore, this type of answer is not sufficient for your assignment, but it shows you how the format should be.*

Appendix

```
library(AER) # load Applied Econometrics with R library
library(quantmod) # Load quantmod library for time series analysis
library(readxl) # Load readxl to load spreadsheet data
library(stats) # Load stats to compute acf functions
```

```

# load US macroeconomic data
library("ggplot2")
library(ggfortify)
library(dplyr)
library(car)
library(tidyverse) # For data manipulation and ggplot2
library(sandwich)
library(forecast)

# load US macroeconomic data
usmac_qt <- read_xlsx("us_macro_quarterly.xlsx",
                     sheet = 1, col_types = c("text", rep("numeric", 9)))

# Fix format of date
usmac_qt$...1 <- as.yearqtr(usmac_qt$...1, format = "%Y:0%q")
# Relabel column names in dataframe
colnames(usmac_qt) <- c("Date", "GDPC96", "JAPAN_IP", "PCECTPI", "GS10",
                       "GS1", "TB3MS", "UNRATE", "EXUSUK", "CPIAUCSL")

# GDP series as xts object
GDP <- xts(usmac_qt$GDPC96, usmac_qt$Date)["1960::2013"]
# GDP growth series as xts object
GDPGrowth <- xts(400 * log(GDP/lag(GDP)))
# Training data
GDPGrowth.train <- na.omit(GDPGrowth["1960::2002"])

# Question 1
plot(as.zoo(GDPGrowth.train), col = "blue", lwd = 2,
     ylab = "Growth Rates", xlab = "Date", main = "U.S. Real GDP Growth Rates")

# Question 2

library(dynlm)
ar1gdp.lm <- dynlm(ts(GDPGrowth.train) ~ L(ts(GDPGrowth.train)))
library(lmtest)
library(sandwich)
coeftest(ar1gdp.lm, vcov. = vcovHAC) # Output w/ robust standard errors

# Output w/ robust standard errors
ar1gdp.Roblm <- coeftest(ar1gdp.lm, vcov. = vcovHAC)
stargazer(ar1gdp.lm, ar1gdp.Roblm, type="text")

```